BLoC (Business Logic Component) pattern is separation of concerns. The app is divided into three main layers:

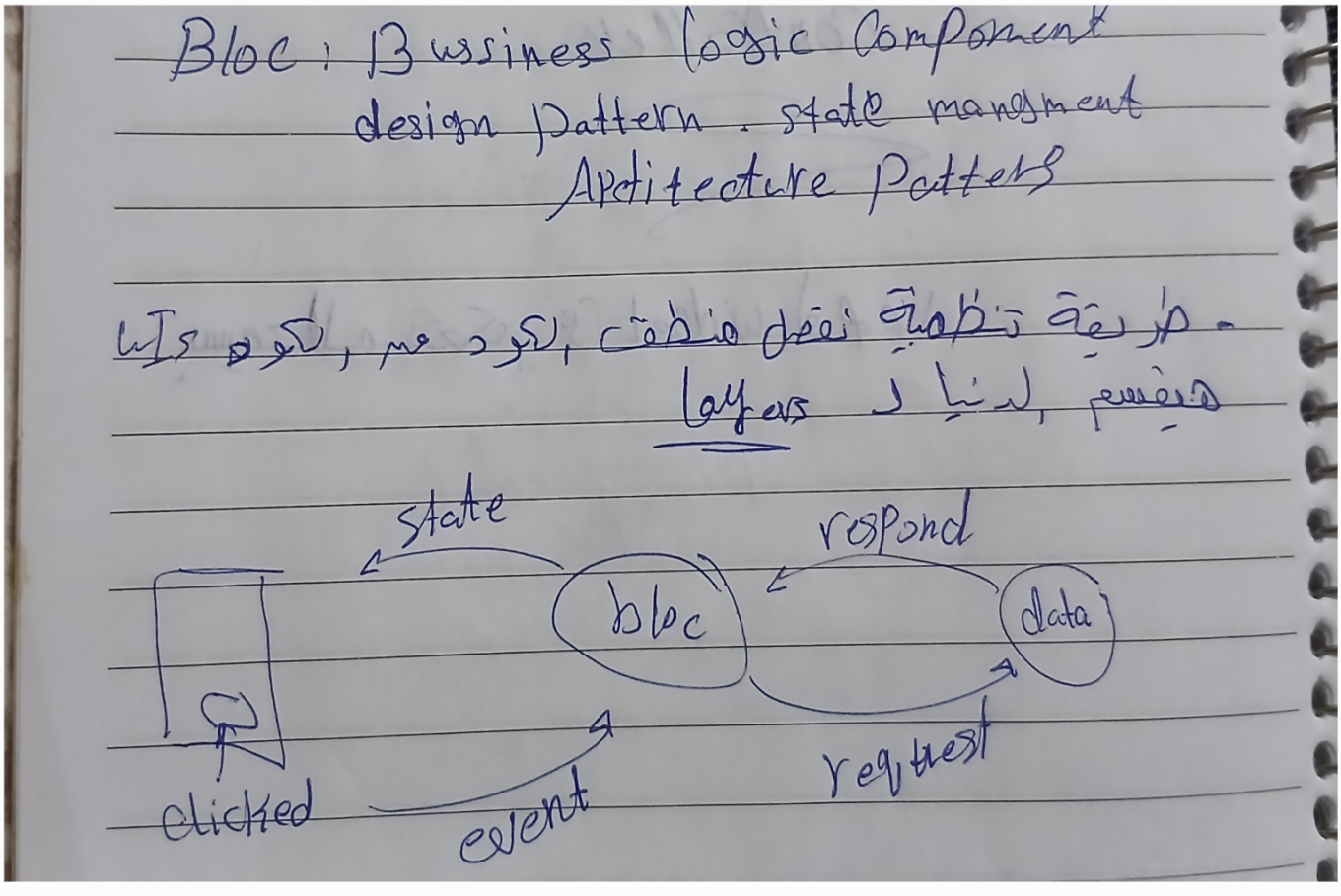
Data Layer:
Responsible for handling data. It includes:
Models (data structure)
Web Services (API calls)
Repository (connects data sources together)

Business Logic Layer:
This is where Cubit or BLoC exists.
It acts as a middle layer between the UI and the data.
It receives input and outputs states.

Presentation Layer (UI):
Displays data to the user and reacts to state changes.

Event:
Represents an action triggered by the user.

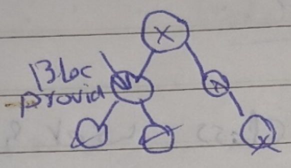
State:
Represents the UI condition at a specific moment

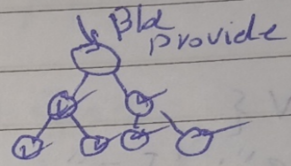


Cubit:
Simpler and easier to use
Does not use events
Directly emits states
Suitable for small or simple features

BLoC:
More structured and scalable
Uses both Events and States
Better for large applications

- BlocProvider → bloc توفر widget
بتديك bloc لكل ال widget تحتها في ال tree





By defult, Bloc provided make the code lazy
* UI gotta call it
BlocProvider.of < -->(Context)

- BlocBuilder → bloc ابعتلي bloc provider بتقول للـ
- rebuild itself w each state

- Bloclistner → بستخدمه في الحاجات ال بتشتغل مره واحده مع ال state change

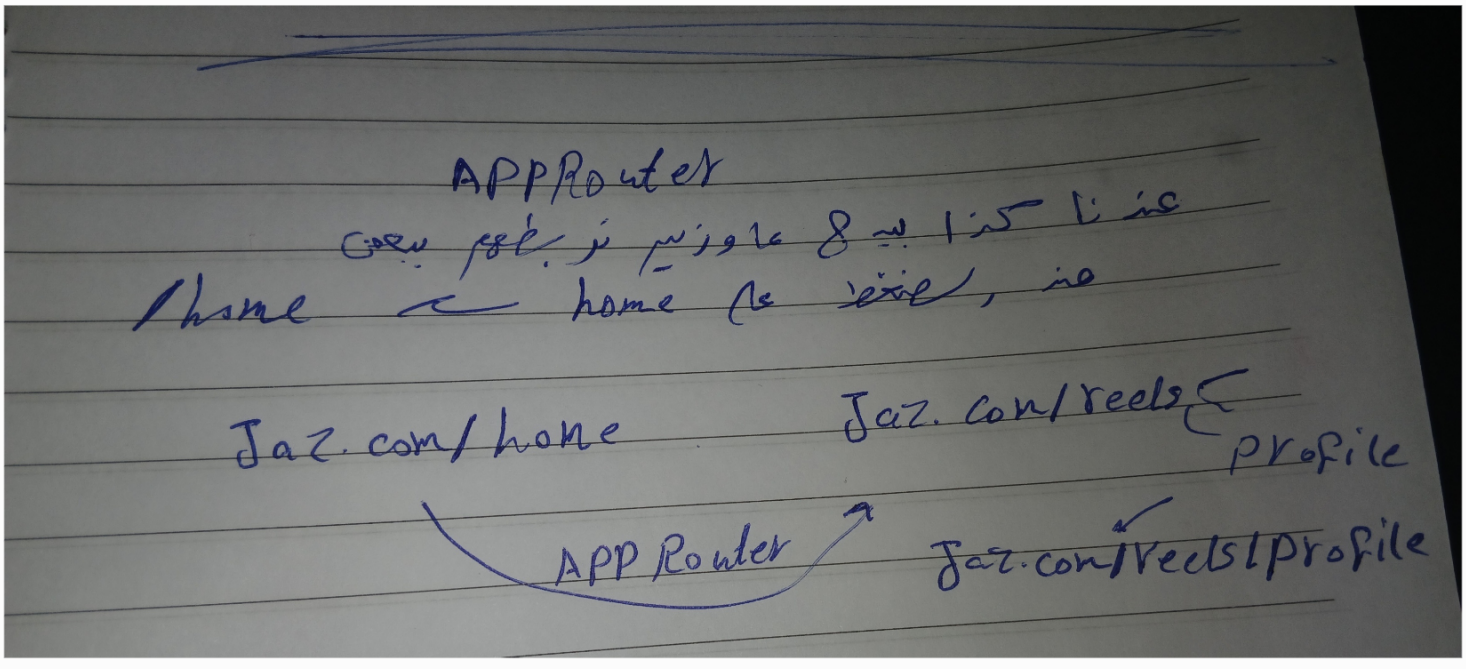
Bloc consumor → BlocBuilder + Bloclistner

Dio:
A powerful library used for making API requests.

JSON to Dart Models:
Converts JSON responses into Dart objects.

Mapping:
Used to convert a list of JSON objects into a list of models using fromJson.

**Instead of random navigation between screens:
We use a centralized AppRouter
Define routes using onGenerateRoute
This makes navigation more organized and controlled



The App Router in Flutter makes navigation between screens more organized and flexible. It helps manage routes in a cleaner way, especially in large applications, and makes the app easier to maintain and scale as more features are added.

Bloc provider in detail

ببساطه هو widget وظيفته يوفر ال Bloc أو ال Cubit لباقي ال widgets اللي جواه في الشجره

```
BlocProvider(
    create: (context) => CounterCubit(),
    child: MyHomePage(), )
```

Create هنا هي بتنشئ ال Bloc أو Cubit وأي widget جوا ال home هتقدر توصل لل Bloc ده

* هو كأننا بنحط object في حته واحده بس وأي widget موجوده تحتيه تقدر توصله وتستخدمه من غير ما تضطر تبعته بإيدك بين الصفحات وال widgets

☺

→ Bloc Builder

ده الجزء اللي بيربط ال UI بال State
أول ما الحاله تتغير، ال widget اللي جوا يعمل rebuild تلقائي ويتحدث بالقيم الجديده

```
BlocBuilder<CounterCubit, int>(
    builder: (context, state) {
    return Text('$state'); }, )
```

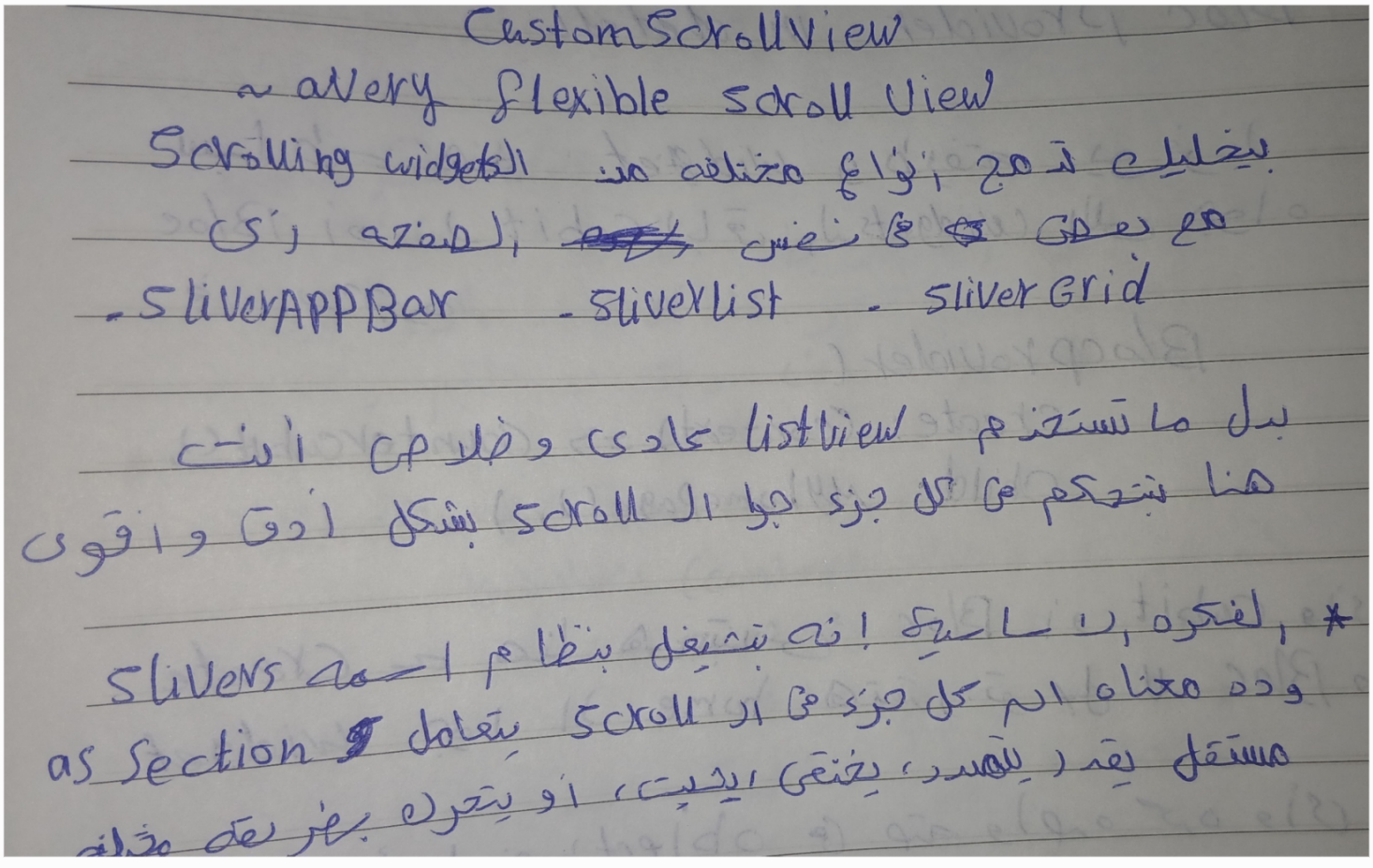
* القيمة الحالية state
* كل ما تتغير، ال Text يتحدث لوحده

Search to Filter in Flutter means filtering a list based on what the user types in the search bar.
As the user enters text, only the matching items are displayed in the UI.
It’s usually done using a TextField, onChanged, and the where() method to update the list dynamically.

```
List<String> names = ["Ahmed", "Ali", "Mona", "Mariam"];

TextField(
  decoration: InputDaCoration(labelText: "Search Names"),
  onChanged: (value) {
    filteredNames = names.where((name) {
      return name.toLowerCase().contains(value.toLowerCase());
    }).toList();

    setState(() {});
  },
)
)
```



CustomScrollView is more powerful when you need:

- a dynamic AppBar
- combining Grid and List in the same scroll
- custom scrolling effects
- multiple different sections within one scroll view

A very simple ex:

```
CustomScrollView(
  slivers: [

    SliverAppBar(
      pinned: true,
      expandedHeight: 200,
      flexibleSpace: FlexibleSpaceBar(
        title: Text("Profile"),
      ),
    ),

    SliverList(
      delegate: SliverChildBuilderDelegate(
        (context, index) {
          return ListTile(
            title: Text("Item $index"),
          );
        },
        childCount: 20, ),)])
```

animated text:
text that changes or moves with animation instead of appearing static. It can be used to make UI more engaging, like typing effects, fading text, or bouncing words.
You usually implement it using widgets like:
AnimatedDefaultTextStyle (for style changes like color, size, weight)
TweenAnimationBuilder (for custom animated transitions)
packages like animated_text_kit for ready-made effects (typing, fading, rotating text)

_____

```
AnimatedDefaultTextStyle(
  duration: Duration(seconds: 1),
  style: TextStyle(fontSize: 30, color:
Colors.blue),
  child: Text("Hello Flutter"),
)
```

وبارك الله فيما رزق.